

The Data Shadow: How Encrypted Traffic Leaks Life

System and Network Security

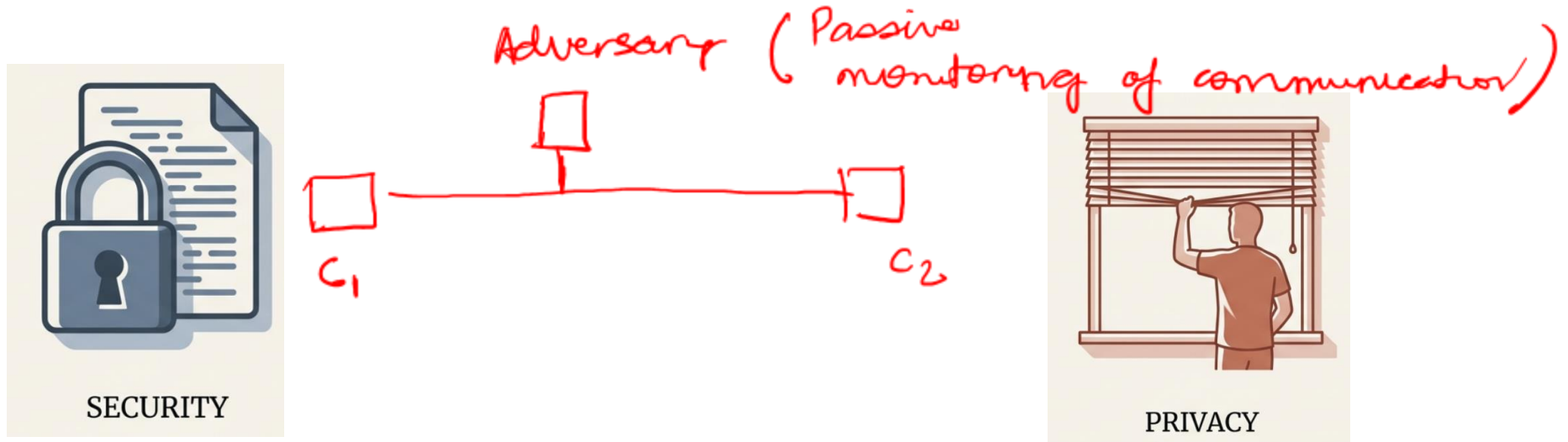
Holi, 2026

Security vs Privacy

If all your Internet communication is encrypted (HTTPS/TLS):

- Are you secure? → encryption, integrity checks
- Are you private?

Threat Model



Security = protecting data from unauthorized access

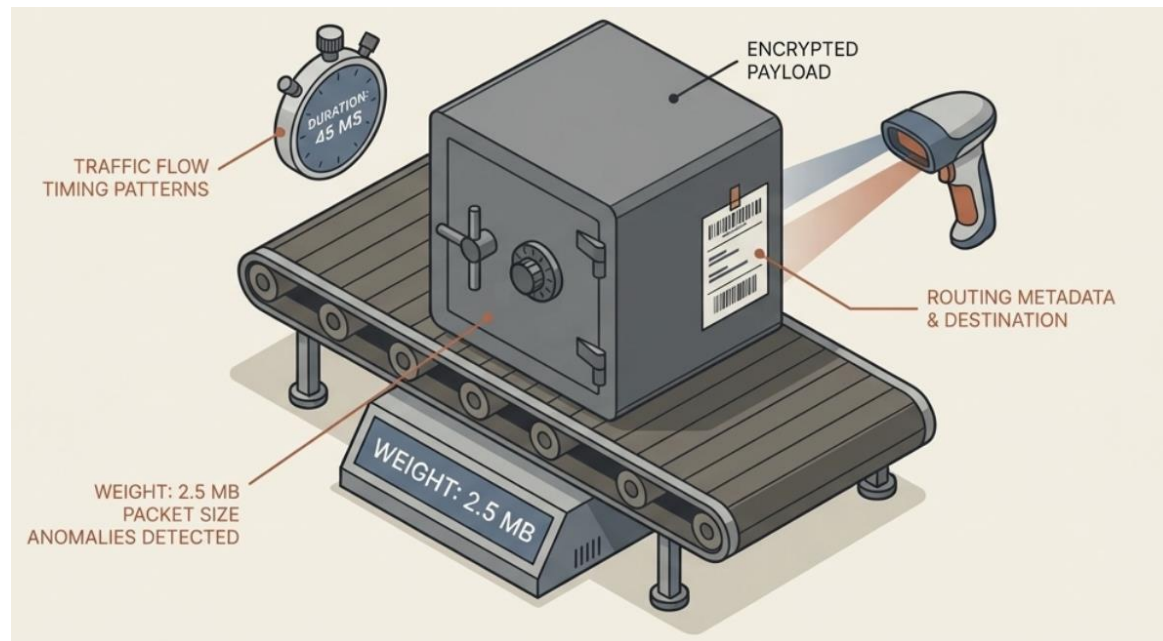
Privacy = controlling what others learn about you

The “Black Box” Problem of Modern Internet Traffic

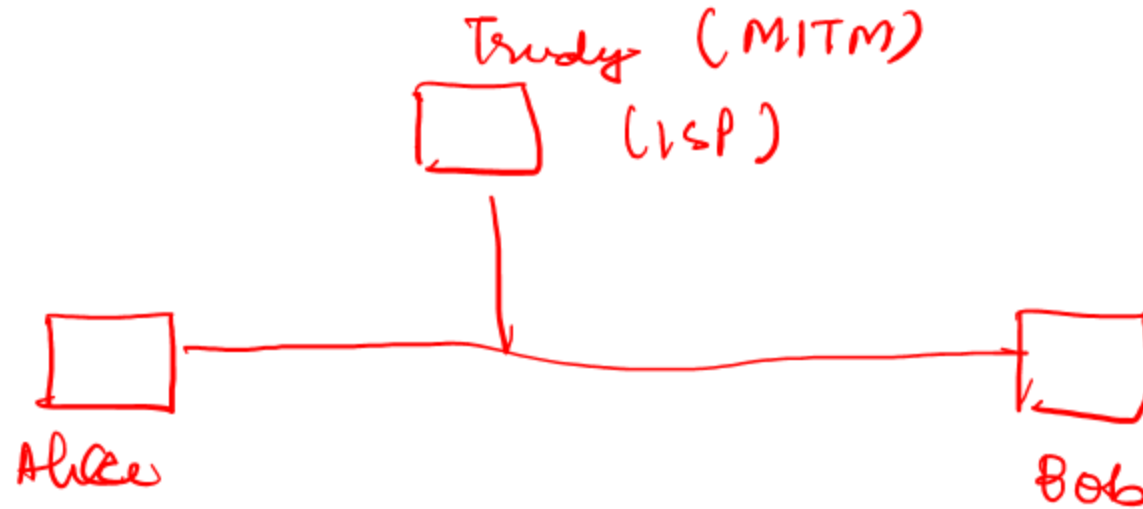
Today, nearly all web traffic uses transport-layer encryption (TLS/SSL). The actual content of your data – the payload – is a black box.

The Flaw: Passive network observers do not need to look inside the box. They simply observe the physical properties of the box itself

Still some
metadata available
to a MITM
↓
Privacy issues



Threat Model



- ① Passive inference of traffic
- ② Traffic is encrypted

Q: What all can be inferred in this case?

What Can An Attacker Observe?

URI: google.com

- Timing of packets
- Size
- Direction
- End-points of packets
- Unencrypted Payload
- MAC address



- ① Packet size
- ② # of pkts
- ③ end points (src - dst)
IP
- ④ Transport protocol
& its headers
- ⑤ Timing of packets
- ⑥ All unencrypted application
payload
 - ↳ DNS
 - ↳ TLS headers (URI)

This Class: Two case studies

- Privacy issues of IoT Traffic
- Privacy issues of MPEG-DASH Traffic

Lecture Organization

- Attacks
- Defense Mechanism

Beauty and the Burst: Identifying Encrypted Video

Dynamic Adaptive Streaming

Over HTTP

- Most video today is streamed using DASH protocol
- Video split into chunks, each chunk encoded into multiple quality levels



- **Claim:** *DASH streams are uniquely bursty* making them susceptible to fingerprint attacks

Why DASH streams are uniquely bursty?

DASH standardizes a leak

- Video is divided into chunks
- Variable Bitrate Encoding: more data for chaotic action scenes and less data for still quiet scenes

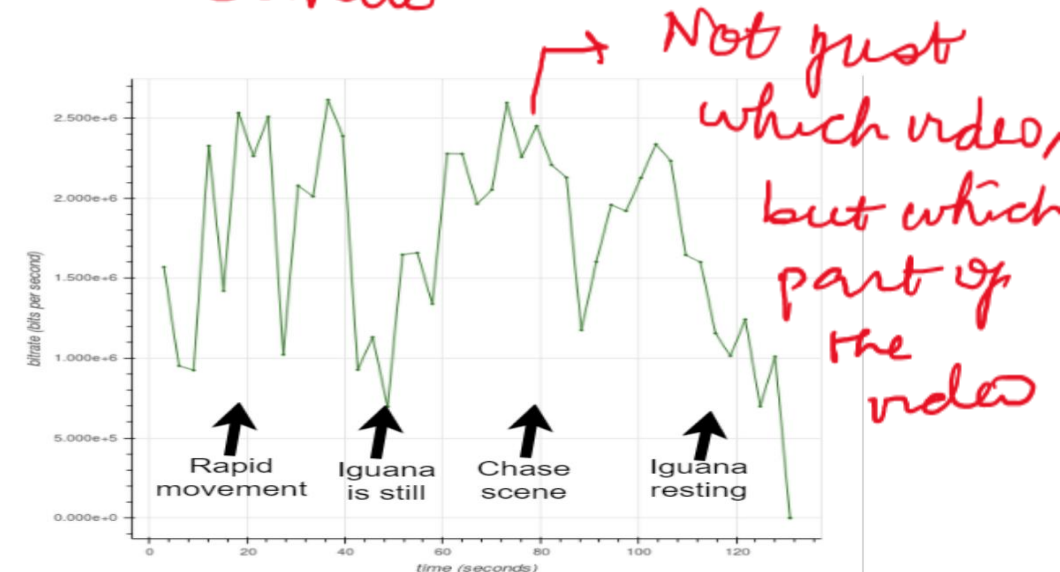
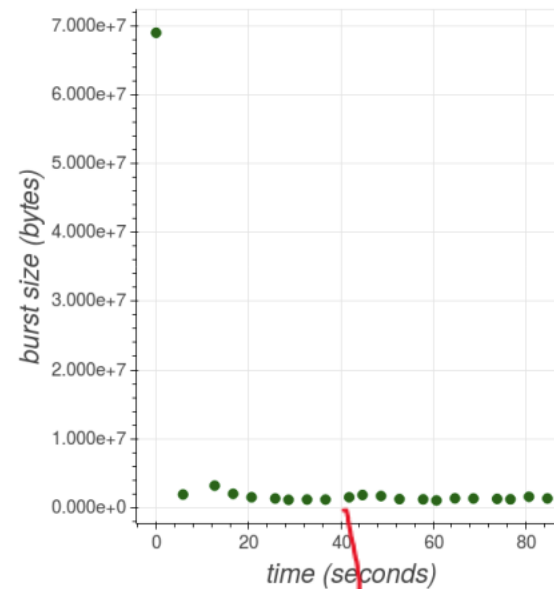
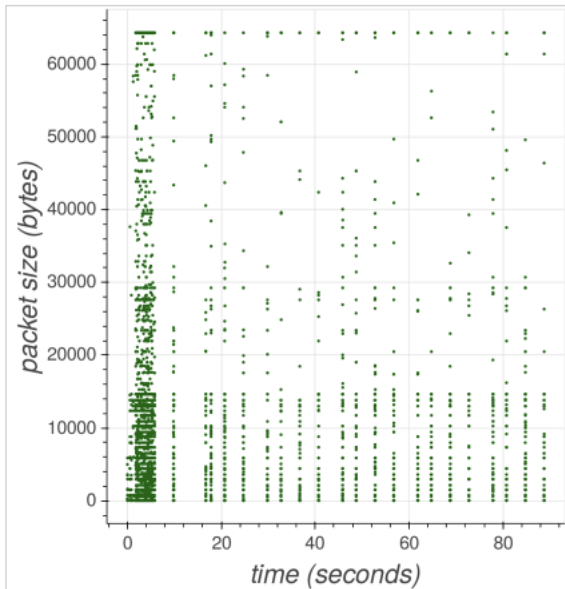
Constant Bitrate Encoding

→ Fixed bitrate, variable

Variable Bitrate Encoding

→ Fixed quality level.

Variable bitrate



→ Fingerprint of the video

Attacker PoV: How To Build The Fingerprint?

Same Burst: if $t_2 - t_1 < \text{Threshold}$
+
interval time

$(v_1, v_2, \dots, v_k) \rightarrow \text{Fingerprint (FP)}$

Burst size

OPTION 2

$(v_1, \dots, v_k, v_2 - v_1, \dots, v_k - v_{k-1}) \rightarrow$

Database of FP to video ID

INFERENCE

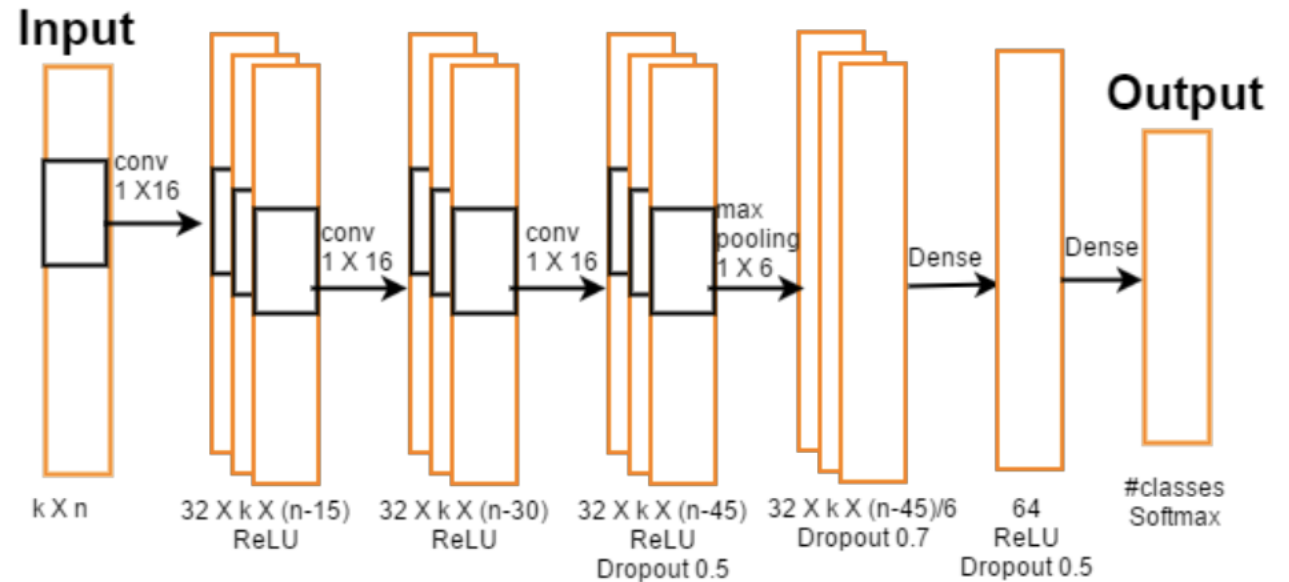
Video X if

$\| \text{Burst Pattern} \|_2$
- FP

$< \text{Threshold}$

Attacker PoV: Real-world Is Noisy

- ① IAT 'can vary' leading to different burst in the n/w
- ② Use ML to mitigate the noise in the network.



A Smart Home is Not A Castle

Flaw: Given IoT devices are single-purpose devices, their network traffic patterns map perfectly to physical human states

→ N/w activity maps to physical activity

Case Study of IoT devices

Device	Network Signature	Behavior Leaked
Sense Sleep Monitor		Exact sleep patterns and times you get out of bed.
Nest Security Cam		Whether you are actively watching the feed, or if physical motion was detected in your empty house.
WeMo Smart Plug		Exactly when physical appliances (lights, coffee makers) are turned on or off.
Amazon Echo		The exact times of day you are home and interacting with a personal assistant.

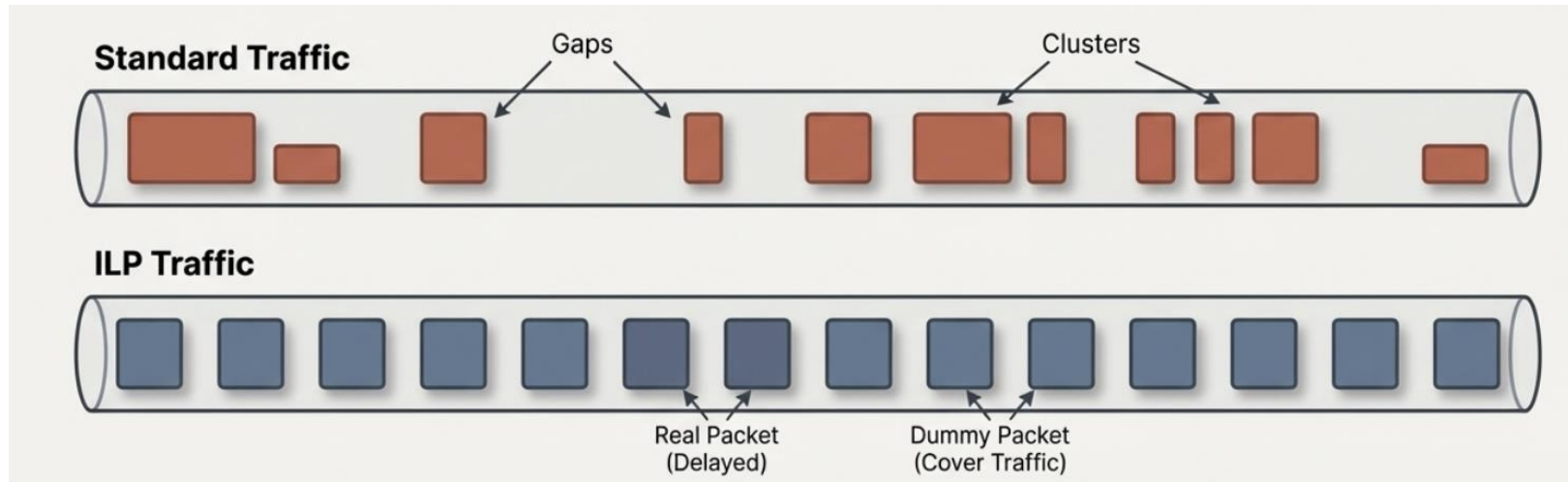
Attack Workflow

- Identify devices inside a home
- Identify video traffic of the device
- Activity inference

One of the 6 Defense: Independent Link Padding

Active state: Shaper delays real packets slightly to fit the transmission rate

Idle State: The shaper injects “cover traffic” to maintain the transmission rate



Summary

- Encryption does not guarantee anonymity
- Metadata can leak behavior
- **Defense:** System-level traffic shaping – adding noise and padding to ensure digital shadows reveal nothing